

This research project consists of **two** assignments. The first one covers chapters 1 and 2, while the second assignment covers chapter 3. The research project will be made available on **March 16th** and should be handed in **no later than June 16th**.

Assignment 1

The file

- Time Series SOI 2010–2025.txt

contains a monthly values time series of Southern Oscillation Index (SOI) from January 2010 to December 2025. Further details are available at <http://www.bom.gov.au/climate/enso/soi/>.

1. Fit a **SARIMA-type** model to the SOI time series. Discuss your model fitting in a step-by-step fashion, presenting your
 - (a) initial examination of the data;
 - (b) transformations (if necessary);
 - (c) initial identification of the dependence orders;
 - (d) parameter estimation;
 - (e) residual diagnostics and model choice.
2. After deciding on appropriate models, forecast the data into the future up to 5 time periods ahead and calculate 95% prediction intervals for each of the 5 forecasts.

Assignment 2

The files

- EDP.xls
- MOTA ENGIL.xls
- GALP ENERGIA-NOM.xls
- THE NAVIGATOR COMP.xls

contain daily share related-prices from four companies (EDP, MOTA ENGIL, GALP ENERGIA-NOM and THE NAVIGATOR COMP) from 2025-02-24 to 2026-01-26, listed on Euronext Lisbon

(<https://live.euronext.com/pt/markets/lisbon>)

namely the date, open, high, low, close, number of shares, number of trades, turnover and vwap.

1. For the four time series calculate its corresponding log-returns associated to the daily closing values;
2. Fit the best **GARCH-type** model to the time series of log-returns. Discuss your model fitting in a step-by-step fashion, presenting your
 - (a) initial examination of the data;
 - (b) initial identification of the dependence orders;
 - (c) parameter estimation;
 - (d) residual diagnostics and model choice.

Grading Criteria for Research Report

- Title accurately reflects content (1.0 point)
- Abstract provides concise summary that highlights most salient points (2.5 points)
- Paper is effectively organized into sections (2.0 points)
- Topic area is covered in sufficient depth (3.0 points)
- The report is statistically/mathematically consistent (4.0 points)
- Tables and Figures are consistent in format (3.0 points)
- Spelling, grammar, presentation (3.0 points)
- List of references (1.5 points)